

Shantanu Shastri

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EDUCATION

Carnegie Mellon University	December 2026
Master of Information Systems Management	GPA: 3.85
Manipal Institute of Technology	August 2022
Bachelor of Electronics and Communications Engineering	GPA: 3.99
<i>Relevant Coursework:</i> Data Structures, Algorithms, Distributed Systems, OOP, System Design	

SKILLS

Languages: Python, TypeScript, Java, C++, JavaScript, SQL, Bash/Shell
Backend & Platform: FastAPI, API Design, Internal Tooling, System Architecture, Node.js, Spring Boot
Data Modeling & Schema: PostgreSQL, Database Design, Data Pipelines, Redis, Neo4j, Milvus, Sybase
Infrastructure & DevOps: Docker, Apache Kafka, Zookeeper, Kubernetes, Azure, AWS, CI/CD, Observability

EXPERIENCE

BlackRock	January 2022 – August 2026
<i>AI Engineer Internship</i>	June 2026 – August 2026

- **Data Pipelines & Integrations:** Built a fault-tolerant Python FastAPI backend and robust data pipelines (Kafka, PostgreSQL) to connect the AI platform with external trade systems for stream-based ingestion.
- **Schema Design & Infrastructure:** Designed clean, well-typed database schemas and infrastructure, linking Milvus embeddings to Neo4j nodes via a PostgreSQL bridge to resolve data inconsistencies.
- **Internal Tooling & APIs:** Engineered internal tooling and a multi-agent platform (RockAI ADK) backed by strictly typed RESTful APIs, enabling the engineering team to safely deploy AI capabilities.
- **Platform Observability:** Built extensive observability workflows to track agent latency and data drift, taking end-to-end ownership to process 3.5M records and handle \$14T in cash.

<i>Software Developer 2 - Associate</i>	January 2025 – July 2025
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- **TypeScript Interfaces:** Engineered well-modeled TypeScript micro-frontends and internal interfaces to translate natural language into JQL, streamlining complex workflows for enterprise users.
- **System Architecture:** Architected scalable Spring Boot data framework and TTL Azure microservices from scratch, managing complex dependencies and reducing infrastructure storage costs by 40%.

<i>Software Developer 1 - Analyst</i>	July 2022 – December 2024
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- **Database Optimization & Modeling:** Took end-to-end ownership of database optimization, refactoring PostgreSQL schemas on tables exceeding 50M+ rows to achieve a 37% reduction in query latency.
- **Foundational Systems:** Built robust backend infrastructure by engineering a sharded server environment with Zookeeper, driving the technical direction to boost system throughput to 10K+ RPS.
- **Rigorous Engineering Standards:** Built comprehensive Azure DevOps CI/CD pipelines with integrated quality gates and JUnit testing suites, ensuring robust, well-documented deployments.

PROJECTS

High-Performance Modular Booking Architecture

- Designed clean database schemas and decoupled microservices using Apache Kafka, seamlessly handling bursts of 100,000+ real-time requests with minimal latency spikes and containerized deployments.

Zulip Open-Source GenAI Extensions

- Built a high-throughput message engine using robust data modeling in Node.js and Redis to process live data streams into concise summaries, prioritizing clean API contracts and zero message loss.